



Ethical Hacking

Questions

- Q1. What are the types of IDS Evidence?
- Q2. What are the tools used in Network Forensics
- Q3. What are the steps in router investigation.
- Q4. What are the sources of Mobile Evidence Collection?
- Q5. Define Mobile Forensic?
- Q6. Enlist the goals of a digital forensic report.
- Q7. Describe the concept of an Intrusion Detection System
- Q8. Demonstrate the steps of mobile forensic investigation in a real-world scenario from evidence collection to reporting
- Q9. Demonstrate how challenges in network forensics can be managed during an investigation
- Q10. Explain the guidelines for digital forensic report writing

ANSWERS

Q1. What are the types of IDS Evidence?

In **Intrusion Detection Systems (IDS)**, evidence refers to the data collected to detect and analyze suspicious activities. The main types are:

- **Alert Evidence**
 - Generated when IDS detects suspicious activity.

- Contains details like source IP, destination IP, time, and type of attack.
- **Session Evidence**
 - Records communication sessions between hosts.
 - Helps reconstruct attacker's actions during a connection.
- **Statistical Evidence**
 - Summarizes traffic patterns and anomalies.
 - Useful for identifying unusual spikes or deviations in network behavior.
- **Packet Evidence**
 - Captures raw network packets.
 - Provides detailed information for forensic analysis of attacks.
- **Log Evidence**
 - System/application logs collected by IDS.
 - Supports correlation with other security events.

Q2. What are the tools used in Network Forensics?

Network Forensics relies on specialized tools to capture, analyze, and investigate network traffic. Common tools include:

- **Wireshark**
 - Popular packet analyzer.
 - Captures and inspects network traffic in detail.
- **Tcpdump**
 - Command-line packet capture tool.
 - Useful for quick traffic analysis and filtering.
- **Snort**
 - Intrusion Detection/Prevention System (IDS/IPS).
 - Detects suspicious patterns in network traffic.

- **NetFlow Analyzer**

- Monitors and analyzes flow-based traffic data.
- Helps identify anomalies and bandwidth misuse.

- **Xplico**

- Network forensic analysis tool.
- Reconstructs application-level data (emails, VoIP, HTTP sessions).

Q3. What are the steps in router investigation.

When conducting **router investigation** in network forensics, the following steps are followed:

- **Identify and Secure Router**

- Ensure the router is isolated to prevent tampering or further compromise.
- Document its physical and network location.

- **Collect Configuration Data**

- Export running and startup configurations.
- Check routing tables, access control lists (ACLs), and firewall rules.

- **Examine Logs**

- Review system logs, event logs, and traffic logs.
- Look for unusual connections, failed login attempts, or suspicious traffic.

- **Check Firmware and Updates**

- Verify firmware integrity and version.
- Look for unauthorized modifications or outdated software.

- **Analyze Traffic Patterns**

- Monitor packet captures and flow data.
- Identify anomalies such as port scanning, DoS attempts, or data exfiltration.

Q4. What are the sources of Mobile Evidence Collection?

In **mobile forensics**, evidence can be collected from multiple sources within a device. Key sources include:

- **Internal Memory (Phone Storage)**
 - Contacts, messages (SMS/MMS), call logs, notes, and application data.
- **External Memory (SD Cards/Removable Media)**
 - Photos, videos, audio files, documents, and cached app data.
- **Communication Data**
 - Emails, instant messages, social media chats, and VoIP records.
- **Network/Service Provider Records**
 - Call detail records (CDRs), SMS logs, and location data from telecom providers.
- **System and Application Logs**
 - OS logs, app usage history, browser history, and GPS tracking information.

Q5. Define Mobile Forensic?

- **Mobile Forensics** is a branch of digital forensics focused on recovering, analyzing, and preserving evidence from mobile devices.
- It involves extracting data such as call logs, SMS, emails, photos, videos, app data, and location information.
- The process ensures evidence is collected in a **legally acceptable manner** without altering or damaging the original data.
- It helps in investigating crimes, cyber incidents, and unauthorized activities involving mobile devices.
- The goal is to provide **authentic, reliable, and admissible evidence** for law enforcement or legal proceedings.

Q6. Enlist the goals of a digital forensic report.

A **digital forensic report** must achieve the following goals:

- **Accuracy:** Present findings clearly and without distortion.
- **Integrity:** Ensure evidence is preserved and reported without alteration.
- **Clarity:** Use simple, structured language so non-technical stakeholders can understand.
- **Objectivity:** Provide unbiased results based only on evidence, not assumptions.
- **Admissibility:** Ensure the report meets legal standards and can be accepted in court.

Q7. Describe the concept of an Intrusion Detection System

- **Definition**
 - An IDS is a security mechanism that monitors network or system activities to detect malicious actions or policy violations.
- **Purpose**
 - Identifies unauthorized access, attacks, or abnormal behavior in real time.
 - Helps protect data and systems by alerting administrators.
- **Types**
 - **Network-based IDS (NIDS):** Monitors traffic across the network.
 - **Host-based IDS (HIDS):** Monitors activities on individual devices.
- **Functionality**
 - Collects and analyzes logs, packets, and system events.
 - Generates alerts when suspicious patterns are detected.
- **Role in Security**

- Acts as an early warning system.
- Supports forensic investigation by providing evidence of intrusions.

Q8. Demonstrate the steps of mobile forensic investigation in a real-world scenario from evidence collection to reporting

A **mobile forensic investigation** typically follows these structured steps:

- **Evidence Collection**

- Seize the mobile device securely.
- Prevent tampering by isolating it (e.g., Faraday bag to block signals).
- Collect SIM card, memory card, and accessories as part of evidence.

- **Preservation**

- Create a forensic image (bit-by-bit copy) of the device's memory.
- Ensure chain of custody documentation to maintain legal integrity.

- **Examination**

- Extract data such as call logs, SMS, emails, photos, videos, app data, and GPS records.
- Use forensic tools (e.g., Cellebrite, Oxygen Forensic Suite) for structured analysis.

- **Analysis**

- Correlate communication records, location data, and app usage.
- Identify suspicious activities (e.g., deleted messages, hidden files, or unusual app behavior).

- **Reporting**

- Document findings in a clear, structured forensic report.
- Include timeline of events, evidence details, and conclusions.
- Ensure the report is legally admissible and understandable by investigators or court.

Q9. Demonstrate how challenges in network forensics can be managed during an investigation

Network forensics faces several challenges, but they can be managed through structured approaches:

- **High Volume of Data**
 - Use automated tools (Wireshark, NetFlow, Snort) to filter and capture only relevant traffic.
- **Encryption of Traffic**
 - Employ decryption techniques or analyze metadata (IP addresses, traffic patterns) when content is hidden.
- **Real-Time Monitoring Needs**
 - Deploy intrusion detection/prevention systems (IDS/IPS) for continuous monitoring and quick alerts.
- **Data Integrity & Chain of Custody**
 - Maintain forensic imaging, hashing, and proper documentation to ensure evidence is legally admissible.
- **Complex Attack Patterns**
 - Apply correlation techniques and anomaly detection to identify multi-stage or hidden attacks.

Q10. Explain the guidelines for digital forensic report writing

A **digital forensic report** must follow clear guidelines to ensure reliability and legal acceptance:

- **Clarity and Simplicity**
 - Write in straightforward language, avoiding technical jargon where possible.
- **Accuracy and Completeness**

- Document all findings precisely, including methods used and evidence collected.
- **Objectivity**
 - Present facts without bias or personal opinions.
- **Structure and Organization**
 - Use a logical format: introduction, methodology, findings, analysis, and conclusion.
- **Legal Admissibility**
 - Ensure chain of custody is maintained and evidence is presented in a way acceptable to courts.
- **Confidentiality**
 - Protect sensitive information and disclose only what is relevant to the case.